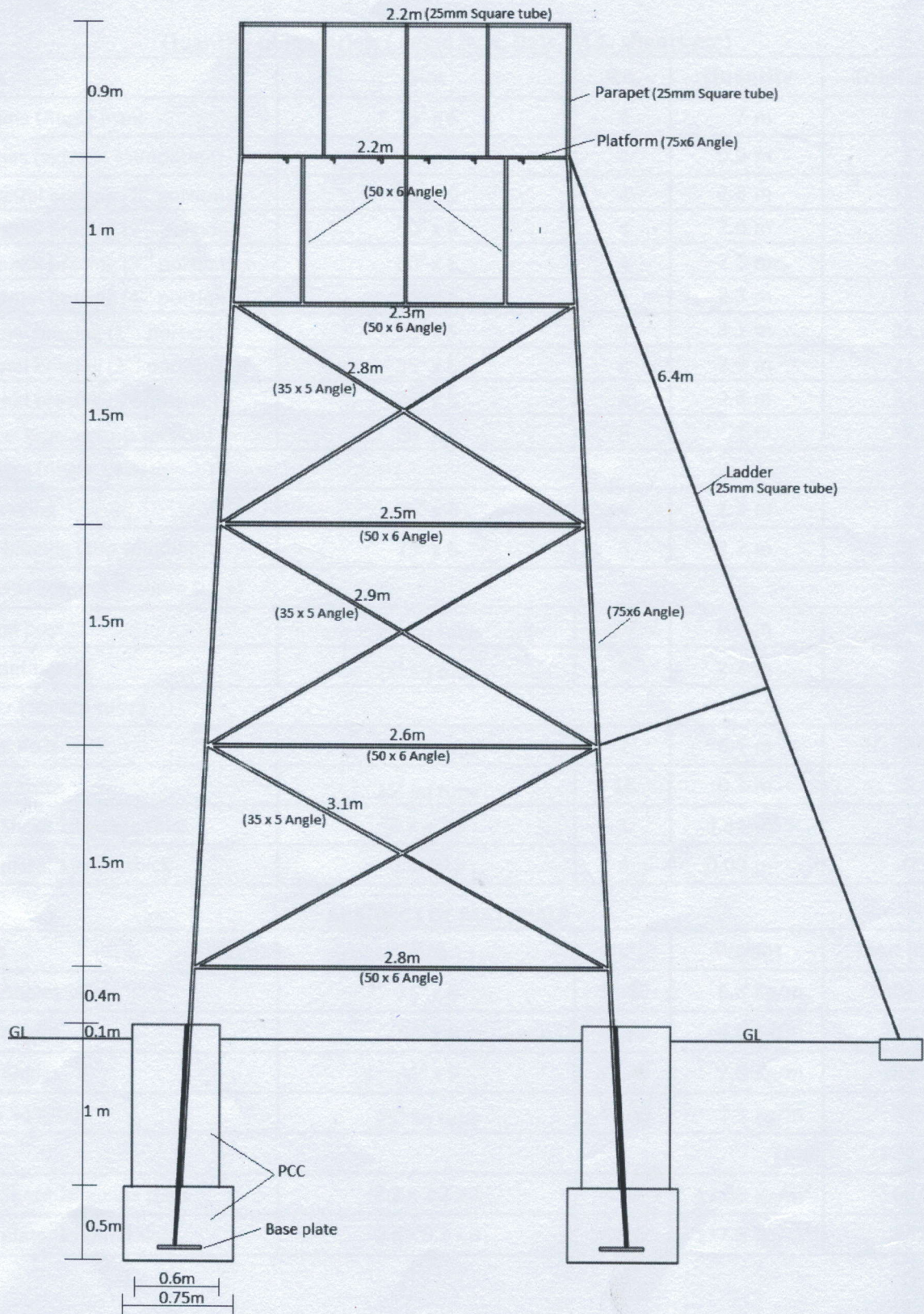


GROUNDWATER DEPARTMENT DISTRICT OFFICE KOTTAYAM

DESIGN OF ANGLE IRON STRUCTURE PLATFORM FOR WATER TANK
CAPACITY - 5000 Ltr, HEIGHT - 6m



District Officer

GROUNDWATER DEPARTMENT DISTRICT OFFICE KOTTAYAM

Estimate For Fabricated Angle Iron Structure To Support 5000 Litres Capacity Water Tank
At 6 m Height

A Quantity of materials (angle iron, flats, M.S. sheets etc)

	Items	Size	No.	Quantity	Total qty
a	Columns (Angle iron)	75 ² x 6	4	7 m	28.00
	Columns (extra in foundation)	75 ² x 6	4	0.9 m	3.60
	Horizontal bracing (1 st portion)	50 ² x 6	4	2.8 m	11.20
	Horizontal bracing (2 nd portion)	50 ² x 6	4	2.6 m	10.40
	Horizontal bracing (3 rd portion)	50 ² x 6	4	2.5 m	10.00
	Horizontal bracing (4 th portion)	50 ² x 6	4	2.3 m	9.20
	Diagonal bracing (1 st portion)	35 ² x 5	8	3.1 m	24.80
	Diagonal bracing (2 nd portion)	35 ² x 5	8	2.9 m	23.20
	Diagonal bracing (3 rd portion)	35 ² x 5	8	2.8 m	22.40
	Vertical bracing (top section)	50 ² x 6	8	1 m	8.00
b	Platform (Angle iron)				
	Top bracing	75 ² x 6	4	2.2 m	8.80
	Cross bracing (Top Middle)	75 ² x 6	6	2.2 m	13.20
c	Platform Parapet (Square tube)				
	Vertical post	25 ² Sq tube	16	0.9 m	14.40
	Horizontal post	25 ² Sq tube	4	2.2 m	8.80
d	Ladder (Square tube)				
	Ladder Post	25 ² Sq tube	2	6.4 m	12.80
	Ladder steps	25 ² Sq tube	18	0.3 m	5.40
e	M. S. Sheet 16 guage thick	2.2 x 2.2	1	4.84 m ²	4.84
f	M. S. plate 15 mm thick	0.3 x 0.3	4	0.09 m ²	0.36

B ABSTRACT OF MATERIALS

	Items	Size	L	Weight	Total (Kg)
1	M. S. Angles	75 ² x 6	53.60	6.8 Kg/m	364.48
	M. S. Angles	50 ² x 6	48.80	4.5 Kg/m	219.60
	M. S. Angles	35 ² x 5	70.40	2.6 Kg/m	183.04
2	25x25 Sq tube	25 ² Sq tube	41.40	1.5 Kg/m	62.10
	Total				829.22
3	M. S. Sheet 16 guage thick	2.2 x 2.2 x 1	4.84	12.56 Kg/m ²	60.79
4	M. S. plate 15 mm thick	0.3 x 0.3 x 4	0.36	117.8 Kg/m ²	42.41

C

ESTIMATE

SI No	Description of work	No	L	B	H	Quantity	Rate	Amount
1	Structural steel work including wastage, cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete as per DSR 10.1							
		1				829.22	75.12 /Kg	62,291.01
2	Steel work including wastage, cutting, hoisting and fixing in position of M. S. Sheet 16 guage thick							
		1				60.79	75.12 /Kg	4,566.57
3	Steel work including wastage, cutting, hoisting and fixing in position of M. S. Plate 15 mm thick							
		1				42.41	75.12 /Kg	3,185.69
4	Providing and fixing bolts including nuts and washers complete as per DSR item 10.20							
		1				5	111.51 /Kg	557.55
5	Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift up to 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m as per DSR item 2.8.1							
		4	0.75	0.75	1.5	3.375	219.02 /m3	739.19
6	Cement concrete 1:2:4, 20mm with nominal size graded stone as per DSR item 4.1.3							
		4	0.75	0.75	0.5	1.125		
		4	0.6	0.6	1.1	1.584		
	Total					2.709	5893.4 /m3	15,965.22
7	Unforeseen items							694.77
	Grand Total Rs.							88,000.00
	Say Rs. 88,000 x Index		Index : 1.3644					1,20,067.00

(Rupees one lakh twenty thousand and sixty seven only)

District officer